

CoCo: Compost Companion

Design and Evaluation of a Wearable Pet That Supports Composting Habits Towards an Interaction Design for Empathy

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ABSTRACT

Food waste is a major source of greenhouse gases and contributes significantly to climate change. Compost Companion, or CoCo, is a wearable anthropomorphic device that uses a GPS system to track composting locations on a college campus to make reducing food waste easy for students. CoCo provides informational feedback on the amount of food that a user has composted or wasted, as well as emotional feedback through the wagging of its tail based on how well the user has been composting. Information including composting locations and amount of food discarded are displayed through an OLED screen, and tail wagging is operated by a Servo motor. CoCo was designed over several iterations based on feedback received from users through surveys, interviews and observations, as well as from subject matter experts through the Delphi method. Overall, findings suggest that CoCo is usable and creates enjoyable composting experiences for users through empathy.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interactive systems and tools**;

KEYWORDS

Anthropomorphic wearable device, climate change awareness wearable, composting habit wearable, compost pet

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1 INTRODUCTION

Food waste is a major contributor of greenhouse gas emissions and is exacerbating climate change worldwide. Each year, 1.2 billion metric tons of food for human consumption are wasted globally, and the United States alone contributes 268 million tons of waste [1, 2]. Moreover, 40% of the food in the US is wasted every year, and 34% of all human-related methane emissions are caused by the

decomposition of food waste in landfills within the United States [3, 4]. Nearly a third of waste in these landfills can be composted, and the impact of every person in the US composting their food waste would be comparable to that of removing 7.8 million cars from the road [5].

To combat the crisis of food waste contributing to climate change, I designed the Compost Companion or “CoCo” (Figure 1). CoCo is a wearable anthropomorphic device for college students that uses a GPS system to track the nearest composting location on campus. The device functions within a system in which compost and landfill bins have built-in GPS trackers and scales to communicate information about location and amount of food wasted or saved to the device (Figure 2).

CoCo can be compared to existing designs for its anthropomorphism and functions. A wearable device in the form of a bracelet developed at MIT Media Lab in 2015 is similar in scale to CoCo and turns everyday objects into game interfaces by providing real-time audio and light feedback [6]. ThermoCaress is a haptic device that uses pressure force and thermal feedback to provide a stroking sensation for the user [7]. Such designs are similar to CoCo in that they provide immediate observable or tangible feedback to the user. However, CoCo goes beyond physical responses to incorporate emotional responses using anthropomorphics.

Chikyuchi and PicoPet use anthropomorphic characters to evoke positive emotions around experiences using games [8, 9]. Chikyuchi uses games specifically related to climate change and green behavior. Contrastingly, CoCo does not reward users based on their performance relative to others in a game, but provides feedback on personal behavior to encourage green habits. Such feedback may be more effective for fostering long-term sustainable behavior, as studies have shown that extrinsic motivation is inversely associated with green habits and can undermine motivation for pro-environmental behavior [10, 11].

Chewpin is a small wearable device that detects eating behavior using acoustic detection of chewing and swallowing [12]. While Chewpin focuses on food consumption itself rather than composting, its technology could be helpful for future improvement of CoCo. For example, detecting the loudness of food dropping could indicate the amount of food being discarded and reduce the communication needed between the compost bin and CoCo.

CoCo’s anthropomorphism functions not only to make composting an experience that is playful and enjoyable, but one that uses users’ empathy for the wellbeing of others to promote sustainable habits. A recent study with socially assistive robots found that users’ facial expressions are a powerful non-verbal measure of empathy [13]. There is a high correlation between feeling empathy and mimicking facial expressions of emotions observed in others [14]. Findings

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Figure 1: The user interacts with CoCo.



Figure 2: CoCo displays the nearest composting location and wags its tail up if the user successfully composts or droops its tail if the user fails to compost.

showed that robots that tell stories in the first-person perspective about fellow robots elicited the greatest empathy from participants.

There is great potential for anthropomorphic devices to elicit empathy from users, which CoCo leverages to specifically aid sustainable habits.

Table 1: Participant responses to interview questions

Interview Question	Participant Responses
How often do you think about climate change?	•“I do think about it a lot, but mostly when it comes to small habits like recycling.” •“I try not to stress over it, but I do think about it when it comes to things like throwing out trash.” •“I do think about it, but I don’t try to dwell on it and feel like I’m in despair.”
Why or why not do you feel like climate change is a pressing issue? How urgent or not urgent is it to you?	•“Climate change is very much real but is often exaggerated and too politicized.” •“Climate change is a very urgent issue and I am pessimistic about our future. The earth could end in 10 years.” •“It is a pressing issue and I do have responsibility, but I feel like big corporations almost have more responsibility and it’s even more urgent for them to change.”
What are some things that contribute to your carbon footprint?	•Driving, using technology, Amazon Prime, leaving the lights on
What are some habits that you think are helping climate change?	•Recycling, re-using water bottles, not purchasing from fast fashion brands, metal straws
Where on your body would you realistically be willing to wear or not wear a wearable device?	•Would wear: socks, necklace, keychains, bow tie, scrunchie, light up shoes •Would not wear: headband, glasses •1 of 5 participants would likely not wear any devices related to promoting climate change and prefers traditional jewelry over wearable devices.

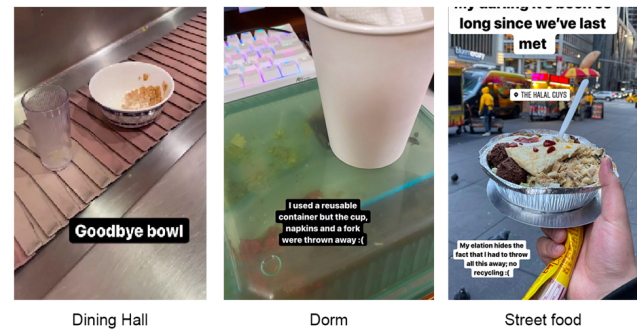
Overall, there are many designs that share similar characteristics or functions with CoCo. However, there are no well-known designs that use an anthropomorphic character to specifically support composting habits through interaction design for empathy.

2 EARLY DESIGN: USER RESEARCH AND PROTOTYPING

I began my design process by choosing to focus on food waste as a contributor to climate change as food waste is an issue influenced greatly by everyday habits. Empathic research highlights the subjective motivations and thoughts that influence people’s behavior and helps design to meet both the physical and emotional needs of users. Through a focus group with 5 participants, I was also able to gather a general understanding of sentiments that college students have around climate change. Overall, participants expressed that they think about climate change mostly in relation to personal habits such as recycling, and have difficulty feeling the impact of their green habits (Table 1). One participant stated that climate change is “real but often exaggerated and too politicized” while another stated that “big corporations almost have more responsibility and it’s more urgent for them to change.”

Based on my interview findings, I identified empathy to be a challenge in motivating students to practice sustainable behavior. Participants struggled to sense the severity and magnitude of harm that climate change brings upon people and the environment due to the lack of immediate personal experiences associated with negative impacts of climate change or positive impacts of sustainable behavior. I came up with the idea of using a wearable pet that would provide emotional feedback to evoke greater empathy around the consequences of environmentally sustainable or unsustainable practices.

To narrow the scope of challenges related to composting, I conducted a cultural probe by asking participants to document a “day in the life” including their food habits through Instagram stories

**Figure 3: Instagram story cultural probe**

(Figure 3). Through this cultural probe, I learned that there are many settings in which users do not have control over whether their food is composted, such as in restaurants, dining halls, and public streets. Consequently, I narrowed the location in which my device functions to buildings on a college campus.

Figure 4 displays a morphological chart used in the design process to make combinations of physical and interactive features to include in CoCo. Outlined boxes highlight features included in the final prototype shown in Figure 2.

Figure 5 displays alternatives to the last prototype shown in Figure 2. The prototype in the left of the figure is a mushroom keychain intended to be hung on a belt or bag. It would display the growth of a plant on an OLED screen to reward the user for composting rather than throwing their food in a landfill bin. The second prototype is an abstract form with a metallic appearance that does not resemble any real plants or animals. This form served as the inspiration for the third form, an earlier version of the final prototype for CoCo. The third prototype displayed an anthropomorphic keychain device intended to be hung on a belt loop or bag. It would display a heart

	1	2	3	4	5	
A	FORM	 mushroom	 flower	 sprout	 abstract	 animal
B	MATERIAL	 fuzzy	 wood	 colorful plastic	 metallic	
C	LOCATION	 city streets	 restaurant	 dining hall	 dorm/apartment	 campus buildings
D	EVENT	 donating food	 eating all food	 composting	 wise shopping	 wise cooking
E	PRINCIPLE FEEDBACK	 LED light flashes	 happy tune plays	 OLED flower grows	 device parts move	

Figure 4: Morphological chart of design variables to implement in CoCo



Figure 5: Alternative prototypes using different combinations of design variables from morphological chart

on an OLED screen to reward user for composting their food rather than throwing it in a landfill bin.

The device was designed to be about 3 inches to be small enough to be comfortably worn by a user but large enough to be seen by others to serve as a conversation starter about climate change. I chose to use an abstract form for the device rather than the form of a real plant or animal to promote the sentiment that one's sustainable habits broadly impact the wellbeing of others, rather than the wellbeing of a particular species.

3 SCENARIO

Debbie is a junior at Cornell double majoring in CS and Economics, and hopes to pursue a minor in Art History. Because of her multidisciplinary studies, Debbie's classes are often scattered around campus so she is not always familiar with the buildings that she finds herself in. Like many juniors, Debbie lives in colleegetown and tries to pack her own lunches. However, she also has the 500 BRB

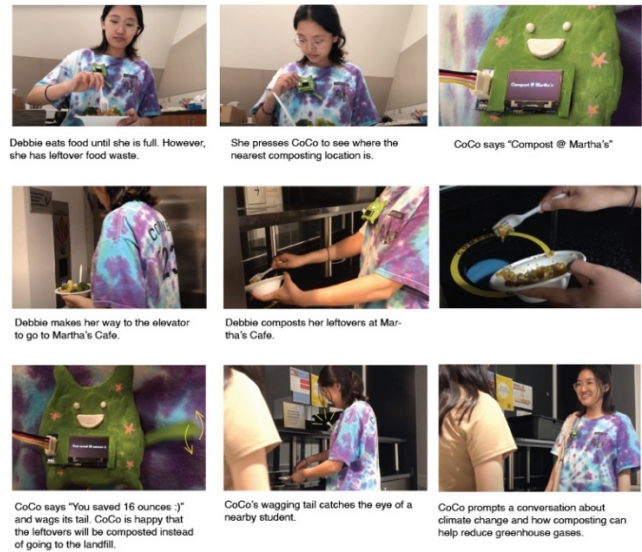


Figure 6: Storyboard of how CoCo helps a user compost and raise climate change awareness

meal plan so she often buys food on campus as well. She stays in the libraries until 10pm, so she eats all her meals on campus.

Debbie feels bad about wasting food, but she is not aware of where to compost on campus and finds it inconvenient to carry around her food scraps all day. Her friend gifted her a Compost Companion or CoCo to help her with this. CoCo is a wearable anthropomorphic device that uses a GPS system to locate and inform users of the nearest composting location on campus.

Debbie presses a button to find the nearest composting location, which CoCo displays on an OLED screen. She presses the button again later to alert CoCo that she has thrown out her food. If Debbie composts, CoCo displays the amount of food she saved and wags its tail to express happiness. If Debbie discards her food in a landfill bin, CoCo displays the amount of food that she has wasted and its tail will droop. The data displayed is the most recent measurement recorded by the scale of the compost bin or landfill bin. Throughout the day, CoCo periodically wags its tail at a different frequency and angle to remind Debbie to compost and inform her of how well she has been composting. A storyboard illustrates the visual sequence of such events (Figure 6).

4 EVALUATION: LOW FIDELITY PROTOTYPE

I conducted a focus group with four students to understand which interactive components users would find most preferable. I offered four potential feedback mechanisms for interaction: color changing LED lights to indicate composting status, playful music as a reward for composting, an OLED screen with facial expressions, and an OLED screen with a flower that grows based on the amount of compost it is fed. All participants stated that they would prefer to not have sound feedback or lights which they perceived to be distracting and childish. They also stated that the OLED screen options would be entertaining, but they would prefer to have quantitative and more tangible feedback on their composting habits.

What would help you to compost most? (1 = extremely unhelpful, 5 = extremely helpful)

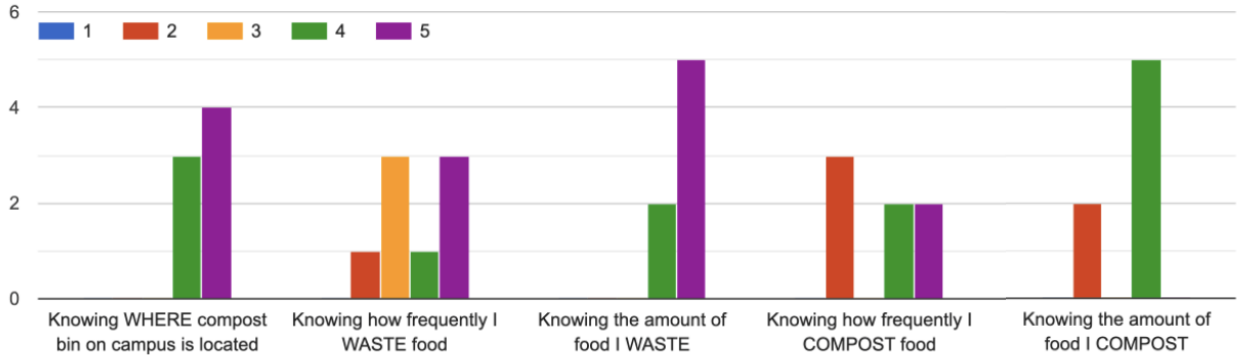


Figure 7: Survey results for preferred information to be displayed by CoCo

How do you feel about potential attributes of keychain that you would wear (1 = strongly dislike, 5 = extremely favorable)

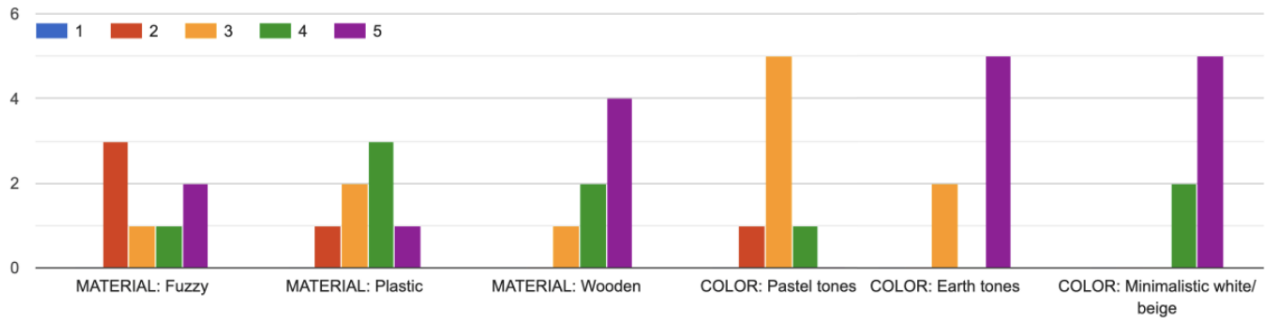


Figure 8: Survey results for preferred materials and visual design choices for CoCo

Next, I conducted a survey with 7 participants to better understand user needs and preferences for materials and functions. 70% of participants reported that they eat on campus every day. However, 75% of participants also reported that they were not well-aware of composting locations and 100% of participants rarely or never compost on campus. Participants indicated that it would be most helpful to know where composting bins on campus are located, and the amount of food that they waste or compost (Figure 7). They also expressed that they would most prefer wooden material and earth tones for the color palette (Figure 8).

The following summarize key design guidelines for CoCo that I gathered from these findings:

1. The device should display the nearest composting location.
2. The device should display quantitative information about the user's composting habits that make the impact of their green habits

more easily understandable. Specifically, the device should display the amount of food wasted and amount of food saved by the user.

3. The device should incorporate earth tones and materials reminiscent of nature.

5 DESIGN: ITERATED PROTOTYPE

The final prototype incorporates a Servo motor and OLED screen to provide feedback on the users' composting habits. The OLED screen displays the name of the nearest composting location, as well as the amount of food saved or wasted by the user. The Servo motor moves a wagging tail which expresses happiness or disappointment based on the user's composting frequency. The speed and angle of the tail wagging convey different emotions. The final prototype is attached to a shirt like a brooch, and can functionally be attached to any fabric surface.

Table 2: SME from the Delphi study agreed that wearing CoCo like a brooch on a shirt would be the most effective locationally.

	SME 1	SME 2	SME 3	SME 4
Round 1	Keychain on belt loop	Brooch on shirt	Brooch on shirt	Band on arm
Round 2	Brooch on shirt	Brooch on shirt	Brooch on shirt	Brooch on shirt

6 EVALUATION: ITERATED PROTOTYPE

I used a standard usability scale questionnaire with 5 participants which yielded an overall usability score of 85.5. I also conducted a Delphi Method study with four subject matter experts (SME) to determine the best location for the device. Originally, the prototype was hung on a keychain and attached to a belt loop. However, only one of four subject matter experts thought that this was the most effective location for the device (Table 2). After revising the prototype to attach to a shirt as a brooch, all subject matter experts agreed that this is the most effective location due to clearer visual access for the user and surrounding viewers.

7 DISCUSSION AND RESULTS

I introduced the Compost Companion or CoCo, a wearable pet that tracks composting locations on the college campus to support personal composting habits. Findings from preliminary studies suggest that CoCo has great usability potential and can motivate sustainable practices in an enjoyable way. Feedback from focus group participants suggests that while users may enjoy playful feedback such as facial expressions, quantitative information on composting habits is greatly desired and perceived as effective feedback for motivating green habits. The standard usability scale questionnaire results suggest CoCo is generally easy to use. However, I also observed that participants experienced some difficulty in reading the OLED screen as the text was upside down for the user. Developing a way to display information effectively for both the user and outside viewers could help make the product more effective overall. Lastly, SME feedback from the Delphi Method study suggests that the product's visibility should be greatly considered in evaluating the effectiveness of its ability to raise awareness around climate change.

8 LIMITATIONS

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the effectiveness of its ability to raise awareness around climate change.

9 CONTRIBUTION AND FUTURE WORK

The design and evaluation of CoCo provides a new perspective in the broader discussion of interaction design for empathy on how anthropomorphism can specifically be utilized to help reduce food waste. Future iterations of CoCo would benefit from finding an effective way to indicate the amount of food discarded, such as through acoustic or motion sensors that would measure the loudness or speed at which food is dropped into a compost bin. Tracking composting locations and the amount of food composted without involving an additional system of GPS trackers and scales to existing compost bins could make the device more readily available to users.

An altruistic interaction design approach could be tested in future iterations to increase social empathy of users. Altruistic interaction design breaks the loop of a closed input-output interaction cycle by making it so that users cannot experience the cycle without the help of others or helping others [15]. Such an approach would involve the composting activity of more than one user to impact the feedback delivered on one device. The involvement of multiple users in the feedback process could instill a greater sense of responsibility in users to compost as their activity would more entities than their own singular device. Further research into effective anthropomorphic emotional responses could also help improve the empathy that is fostered around climate change through CoCo. For example, different body language such as tail wagging or ear wagging could have varying degrees of effectiveness in communicating an emotional response to users.

CoCo uses empathy through anthropomorphism to encourage users to practice sustainable habits. Designing for empathy can be a powerful way to combat the climate crisis through changes in personal behavior and awareness around the crisis. Moreover, experiences for users that are not only playful but also informative provides them with a sense of value that encourages them to practice sustainable behavior.

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